

MCP Semantic Annotations for EE2 Compliance

Presentation Figures and Charts

NOAA/EMC Environmental Information Branch (EIB)

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1 Architecture Overview

v7.0.0 Single Source of Truth Architecture

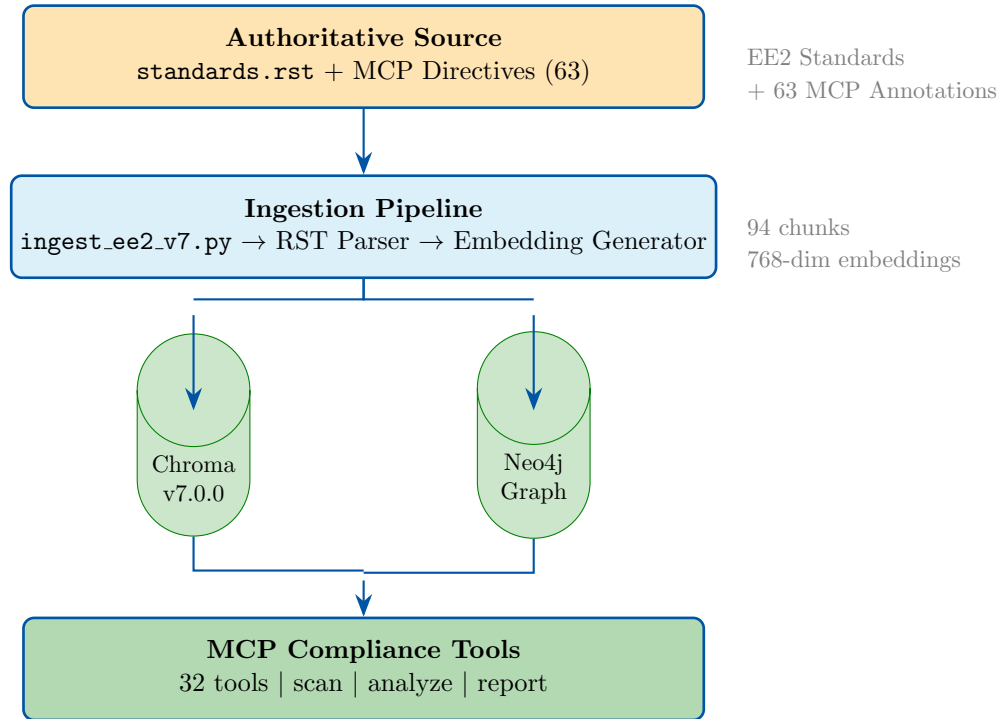


Figure 1: MCP Semantic Annotation Architecture (v7.0.0)

2 Embedding Pipeline Flow

Embedding Convolution Pipeline

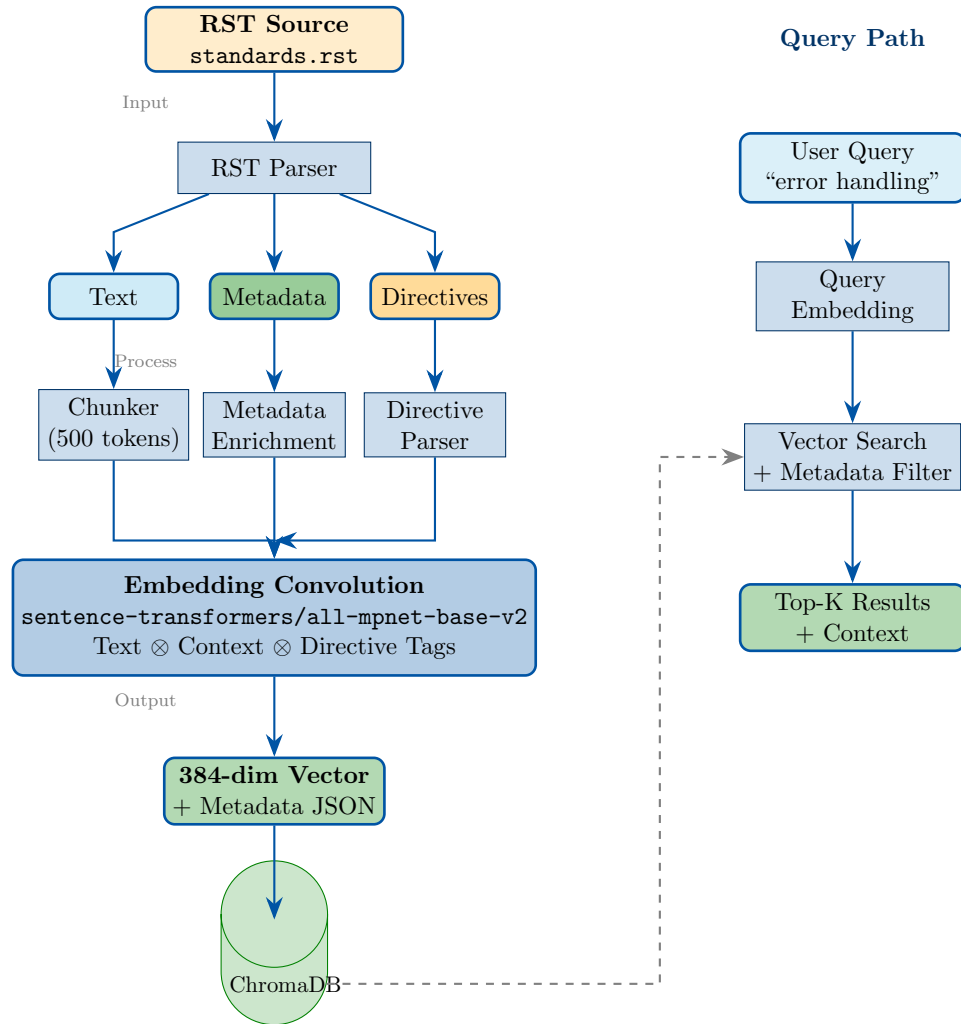


Figure 2: Embedding Convolution: How RST Directives Become Searchable Vectors

3 False Positive Reduction Results

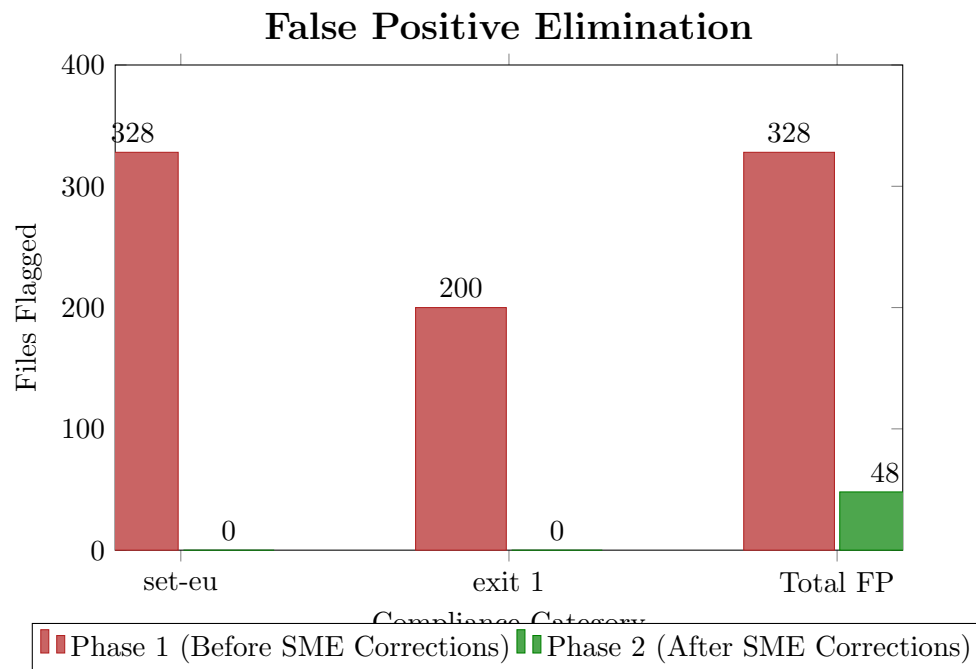


Figure 3: 85% Reduction in False Positives Through Semantic Annotations

4 MCP Directive Distribution

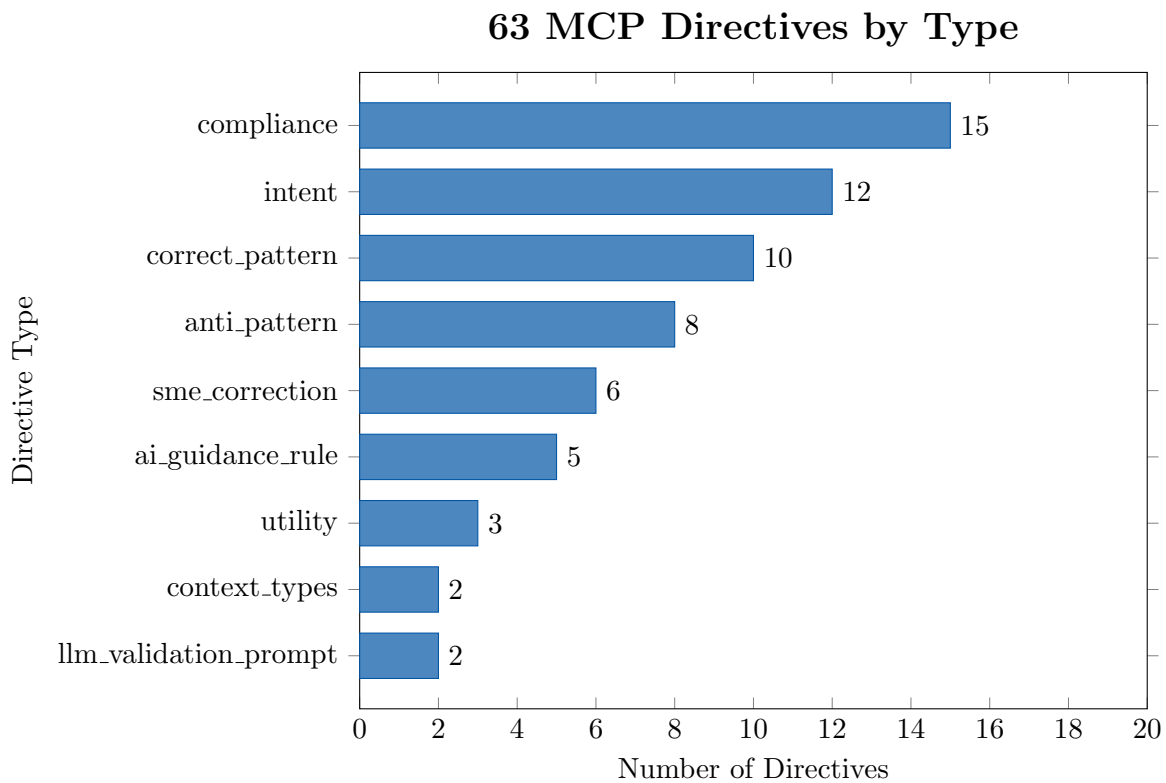


Figure 4: Distribution of MCP Directive Types in standards.rst

5 EE2 Category Coverage

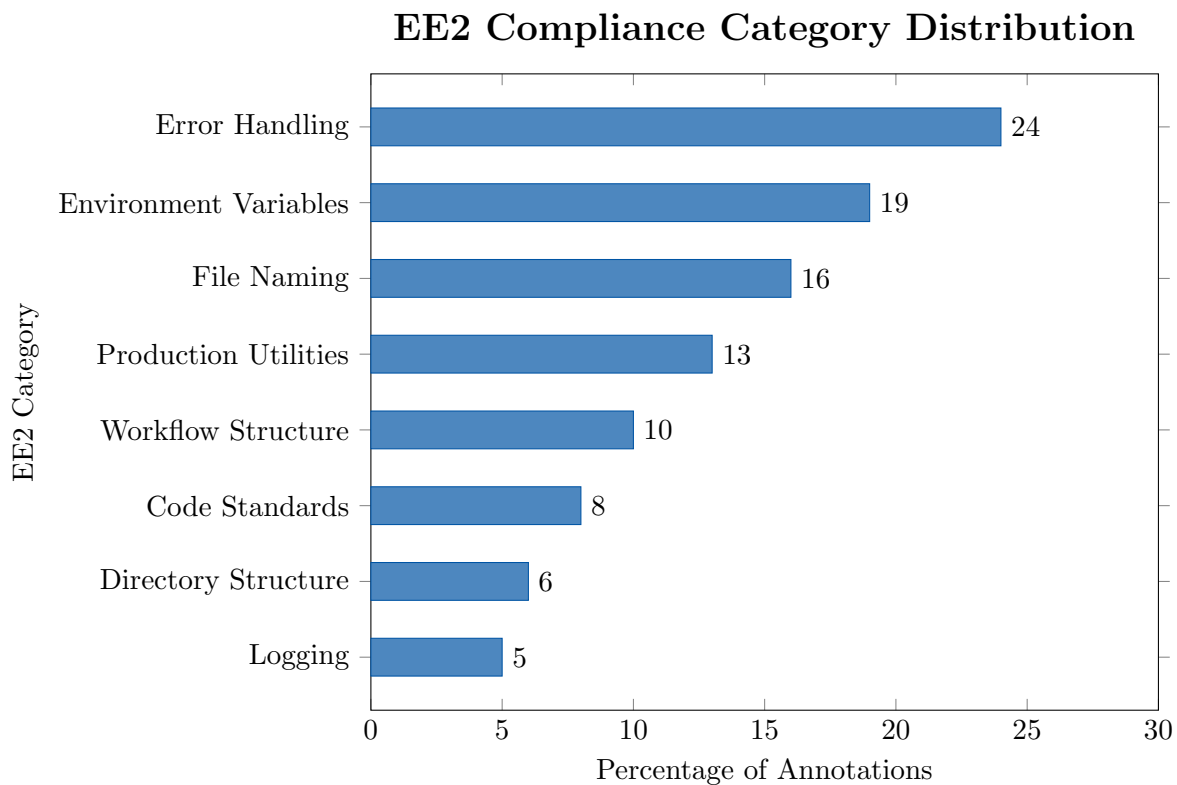


Figure 5: Distribution of Annotations Across EE2 Compliance Categories (63 total directives)

6 System Component Diagram

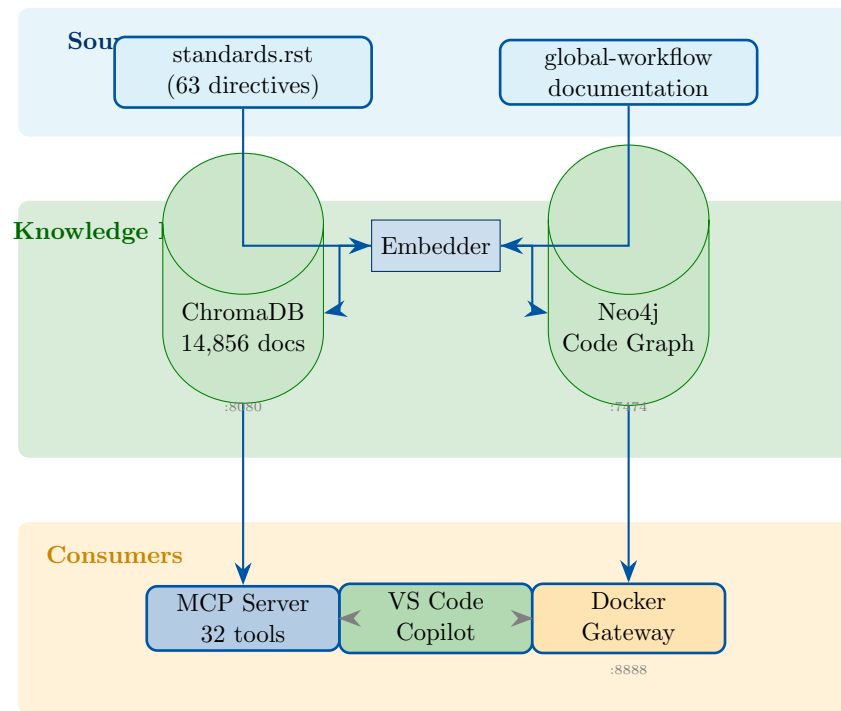


Figure 6: MCP-RAG System Components and Data Flow

7 Update Workflow

Zero-Code Rule Update Workflow

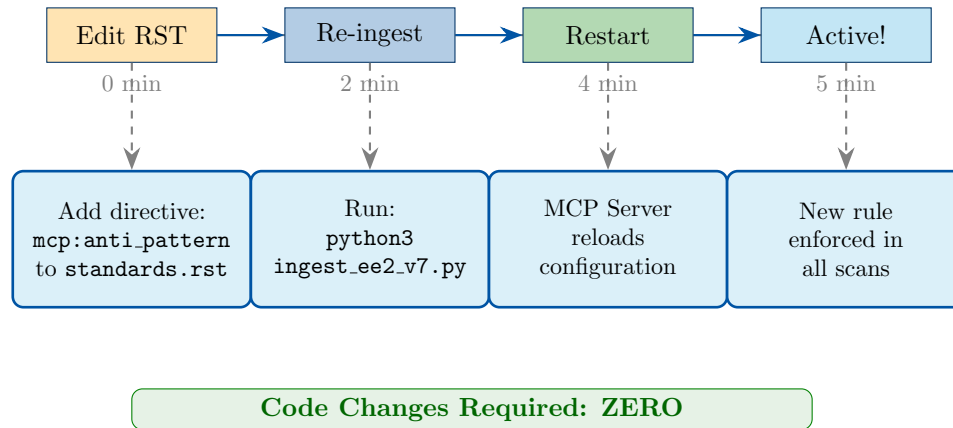


Figure 7: 5-Minute Update Workflow with Zero Code Changes

8 Before/After Comparison

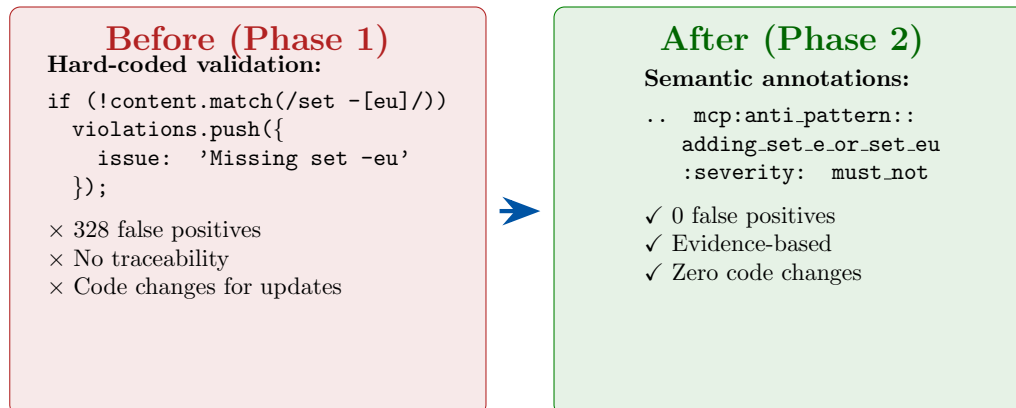


Figure 8: Transformation from Hard-Coded to Semantic Validation

9 Summary Table

Table 1: MCP Semantic Annotations System Metrics

Metric	Value	Notes
<i>Knowledge Base</i>		
ChromaDB Collections	12	v7.0.0 architecture
Total Documents	14,856	Across all collections
EE2 Chunks	94	standards.rst content
MCP Directives	63	Inline in standards.rst
Embedding Dimensions	768	all-mpnet-base-v2 model
<i>Compliance Impact</i>		
False Positive Reduction	85%	328 → 48 violations
set -eu FP Eliminated	100%	328 → 0 files
exit 1 FP Eliminated	100%	200 → 0 files
<i>Operational</i>		
MCP Tools Available	32	7 tool modules
Rule Update Time	5 min	Zero code changes
Config Load Time	100ms	At server startup
Scan Speed	18ms/file	No runtime DB queries

10 Appendix: Compiling This Document

To compile this document:

```
# Local compilation
pdflatex mcp_semantic_annotations_figures.tex
pdflatex mcp_semantic_annotations_figures.tex # Run twice for ToC

# Or use latexmk
latexmk -pdf mcp_semantic_annotations_figures.tex

# Required packages:
# - tikz, pgfplots
# - booktabs, float
```

For Overleaf: Upload this file directly and compile.